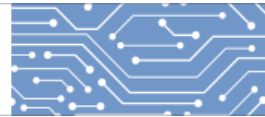


Americas Technology: Semiconductors: Communacopia + Technology Conference 2026 – Day 2 Takeaways

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Companies from the US Semiconductor sector presented during Day 2 of the GS Communacopia + Technology Conference, including TSMC, Amkor, NXP Semiconductors, Cadence, SanDisk, IBM, Teradyne, Applied Materials, and SiTime. **We highlight key takeaways on the top debates, along with company-specific takeaways for each sector below.**

Takeaways for US Semiconductors

Key Takeaway #1 – AI Adoption Trends and EDA Software: Companies discussed how agentic AI is driving significant demand tailwinds for both services and equipment, all the way from systems to components. Importantly, a number of companies cited concrete ways in which they are implementing AI in their internal business processes to **drive incremental revenue growth** via new product development and faster innovation cycles in services. Companies also cited their use of internal AI tools to drive cost efficiencies, in line with past commentary.

- **Cadence:** Cadence continues to see design momentum with traditional and non-traditional chip and system designers. The company noted chip complexity is expected to grow by 48X over the next five years, which is a key driver for its agentic AI solutions – enabling accelerated recurring revenue growth. The company also shared optimism around physical AI, and expects it to be a large market opportunity in the medium-term.
- **IBM:** IBM highlighted the benefits of its cost-effective, open-source AI tools such as its smaller LLMs, RHEL AI, WatsonX code generation tools, and WatsonX Orchestrate for scaling agentic AI. Management also highlighted the growing importance of customer's data in the context of AI, and sees GenAI as a key new source of value in its Consulting business.
- **Applied Materials:** Applied noted shorter time to market, and yield and output innovation driven by AI as key ROI provider in the industry to Applied and to its customers. AI is increasingly being applied to new product development that

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translates to incremental revenue growth and market share gains for the company over time. Applied also cited improved value proposition in services using AI as an additional driver of ROI in the industry.

Key Takeaway #2 – Semiconductor Capital Equipment, Foundry and Memory:

Companies highlighted higher structural demand driven by AI, which are driving capacity additions given extended demand visibility from customers. Supply/demand remains tight across the supply chain, as evidenced by comments from TSMC (relative to advanced packaging and wafer capacity), Amkor (relative to advanced packaging) and Teradyne (relative to testing). Specific to memory, SanDisk noted that NAND remains under tight supply/demand conditions – and this is expected to last through 2027 and potentially 2028.

- **TSMC:** TSMC continues to view AI as a multi-year megatrend that remains at a very early stage, with lower compute costs expected to accelerate AI adoption and ultimately drive greater demand for advanced silicon. TSMC noted that execution in Arizona continues to progress well, with Phase 1 achieving yields comparable to Taiwan and already generating profits, while TSMC expects approximately 30% of its N2-and-beyond capacity to be located outside Taiwan in five years, primarily in Arizona. Despite dilution from the N2 ramp and overseas expansion, management remains comfortable achieving its long-term gross margin target of 56% and above, supported by disciplined capacity planning, high utilization, productivity improvements, cost reductions, and pricing that reflects TSMC's value.
- **Amkor:** Management discussed phase 2 of its Arizona campus, driven by strong demand for advanced packaging. Amkor expects compute revenue to continue to grow at 30% YoY in 2027. Amkor reiterated its \$11 bn 2030 revenue target, and expects 2028 CapEx levels to be elevated and sustain in 2029, before moderating in 2030 and 2031.
- **SanDisk:** The company remains confident in its long-term financial targets underpinned by its long-term agreements (NBMs), which themselves have been enabled in part by strong and growing datacenter demand. Management also highlighted its low capital intensity, and reiterated it sees share repurchases as central to its capital return policy.
- **Teradyne:** Teradyne sees increasing opportunities with hyperscale customers, whose diverse requirements across chip, system, and datacenter testing create reinforcing demand across all three businesses. Management also expects the test equipment market to at least grow in line with overall semiconductor CapEx. Teradyne expects AI-driven growth to accelerate through 2027, supported by rising share gains at a major merchant compute customer, expanding engagements with hyperscale custom silicon customers, and strong momentum in ASIC testing.
- **Applied Materials:** Applied Materials expects leading-edge logic, advanced packaging, and memory to account for ~80% of WFE in 2026 and similar mix going ahead, and also reiterated its outlook for ~40% revenue growth in 2026. Management sees strong growth continuing into 2027 and 2028 given increasing orders and backlog with longer-dated visibility. The company also expressed confidence in its Process Control business, which is expected to grow more than 50%

in 2026, driven by increasing demand for yield improvement, performance optimization, and materials engineering at advanced nodes.

Key Takeaway #3 - Analog: Companies struck a bullish tone on a broad-based recovery across end markets - including industrial and automotive - driven by lean inventory levels, content growth, and steadily improving demand. Demand growth remains outsized in idiosyncratic end markets such as datacenter and aerospace/defense - and this is expected to continue in these areas for the foreseeable future.

- **NXP:** Management highlighted improving demand trends across end markets, including in automotive, and improving revenue visibility through longer lead times despite a lack of OEM inventory restocking. NXP also discussed its strategic pivot to physical AI, its hybrid manufacturing strategy, and its gradual price increases and their impact on revenue and margins.
- **SiTime:** The company discussed its constructive outlook for datacenter, which remains its key growth driver, driven by AI server units, increasing market share, and higher content. SiTime also noted that its Renesas IC clock revenue growth is expected to inflect in 2027 and that overall revenue growth is likely to exceed its long term targets over the next few quarters.

Key Takeaway #4 - IT Services: IBM expects to see accelerating revenue growth in its Consulting business in 2H26, which it believes is idiosyncratic in nature due to its larger exposure to Gen AI service areas relative to peers.

- **IBM:** Following its recent guide down, IBM is on track to deliver 4% - 5% revenue growth in CC for 2026, while increasing free cash flow by ~\$1bn YoY. In Software, a shift in client spending priorities had driven a shortfall in Transaction Processing and Automation - but IBM expects to grow its Software segment 6% - 8% in 2026, led by Red Hat and momentum across its recurring revenue business. IBM sees GenAI as a key source of value for its consulting business going forward, and expects to drive accelerating Consulting revenue growth in 2H26. Management believes that quantum computing could represent nearly \$1bn of revenue for IBM before 2030.

Key Takeaways from TSMC presentation

Presenters: Wendell Huang, SVP & CFO

- **AI remains in the early stages of a multi-year megatrend:** TSMC continues to see increasing adoption of AI. Management also highlighted the shift from generative AI toward agentic AI, which should broaden compute demand beyond accelerators to CPUs. TSMC remains confident in the durability of the AI cycle based on discussions not only with its direct customers but also with its customers' customers. Recent visits of US CSPs further reinforced management's confidence, as cloud service providers appeared well prepared in areas including power and broader infrastructure deployment.
- **Lower compute costs should drive greater consumption and silicon demand:** Management believes open-source models and continued improvements in

computing efficiency will reduce token costs, leading to greater AI consumption and broader adoption, supporting incremental demand for advanced silicon. TSMC has already incorporated this dynamic into its capacity planning and continues to expand aggressively to support what it views as structural, multi-year demand growth.

- **Disciplined demand validation remains critical amid aggressive expansion:** TSMC uses a combination of top-down and bottom-up analysis to validate demand. Its top-down assessment considers macroeconomic conditions, semiconductor demand, fabless/system companies, the foundry industry, etc, while its bottom-up approach evaluates demand across different segments and incorporates feedback from customers and customers' customers. Management acknowledged that semiconductors will remain cyclical, but believes correctly identifying the underlying megatrend is more important than attempting to forecast every short-term fluctuation.
- **Arizona execution is demonstrating that TSMC's manufacturing model can travel:** TSMC reiterated that overseas expansion is based on customer demand for geographic flexibility and an appropriate level of government support. Arizona Phase 1, using N4 technology, is in operation, with yields comparable to Taiwan, satisfactory quality and reliability, and positive profitability. Management views this as an important milestone because it demonstrates internally and to customers that TSMC can produce wafers in the US with the same quality as in Taiwan. Equipment move-in for Phase 2, which will use N3 technology, is expected to begin soon; construction of Phase 3 has started; and preparation work for Phase 4 and the first US advanced-packaging facility is underway.
- **Arizona has substantial room for further expansion:** In addition to its original Arizona development, TSMC has acquired another parcel of land capable of accommodating an additional five to six fabs, supporting its recently announced incremental US\$100bn investment. Based on its current plan, TSMC expects approximately 30% of its N2-and-beyond capacity to be located outside Taiwan in five years, primarily in Arizona. However, the most advanced technologies will continue to ramp first in Taiwan because the early-stage technology transfer requires close collaboration and frequent movement of engineers among R&D and manufacturing sites. Overseas deployment will generally follow after the technology has stabilized in Taiwan.
- **Customers increasingly value and are willing to pay for geographic flexibility:** Management noted that customers initially requested only a modest amount of geographically diversified supply, but their requirements have increased as TSMC has demonstrated successful execution in Arizona. TSMC believes this geographic flexibility represents incremental value and intends to reflect that value in its pricing, with customers increasingly willing to recognize the associated premium.
- **Technology, manufacturing excellence, capacity, and trust underpin TSMC's competitive moat:** Management believes customers select a foundry partner based on access to the best technology, the ability to ramp rapidly into high-volume manufacturing with attractive economics, sufficient capacity, and confidence that

the foundry will not compete with its customers. TSMC is confident in maintaining this lead by continuing to advance its roadmap. Management also highlighted its record of rapidly ramping new technologies, the scale of its wafer and advanced-packaging investments, and its consistent delivery against customer commitments as key elements of customer trust that cannot be replicated through CapEx or engineering recruitment alone.

- **Pricing remains strategic rather than opportunistic:** TSMC does not intend to raise prices simply because near-term utilization is high, nor reduce prices simply because utilization temporarily weakens. Instead, the company's approach is to price sustainably and consistently based on the value it provides, while ensuring that customers can remain successful over the long term. Management noted that TSMC's structural profitability has continued to improve, with its long-term gross margin target increasing from 50% five years ago to 53% and now 56% and above.
- **Leading-edge wafer and advanced-packaging capacity remain tight:** Management continues to build capacity wherever possible to meet customer demand. N3 demand remains particularly strong: unlike prior technology cycles, when demand for a node typically began declining two to three years after its introduction, TSMC is adding further N3 capacity and still does not expect supply to be sufficient. The company is also expanding advanced-packaging capacity to address strong accelerator demand. Outside TSMC, management identified power and memory as potential bottlenecks cited by other parts of the AI supply chain, while emphasizing that foundry capacity itself remains difficult to expand because fabs require two to three years to build and another one to two years to ramp.
- **TSMC's disclosed AI revenue understates its broader exposure:** Management reiterated that its disclosed AI revenue definition is relatively narrow and primarily captures GPUs, custom AI accelerators and memory base die. It excludes CPUs and networking products because TSMC cannot reliably determine whether each chip is ultimately deployed in an AI system. As agentic AI drives incremental demand across x86, Arm, and RISC-V CPUs, as well as networking and other supporting silicon, TSMC's total AI-related exposure is therefore broader than its reported AI-accelerator revenue. The company intends to retain its existing definition until it can more accurately distinguish the end use of these products.
- **No visible barrier to continued transistor scaling:** TSMC is currently ramping N2, with A14 expected approximately two years later, while development of the subsequent node is already underway. Management does not currently see a technical issue preventing delivery of the next node or continued advancement beyond the existing roadmap. TSMC views close collaboration with customers as a key foundry advantage, allowing the company to develop each technology against specific customer performance, power, density, and economic requirements.
- **Cross-node capacity optimization is an underappreciated advantage:** TSMC highlighted its ability to generate additional output through productivity improvements and to convert tools and capacity across technology nodes. For example, the company can use portions of its N5 capacity to support N3 demand and can also optimize capacity across less-adjacent nodes where technically

feasible. Management views this cross-node flexibility as a distinctive element of TSMC's manufacturing excellence, allowing the company to respond more quickly to demand changes while reducing the risk of capacity obsolescence.

- **Advanced packaging is strategic primarily because it enables wafer growth:** CoWoS capacity remains tight, and TSMC already started outsourcing selected packaging processes to OSAT partners to support additional growth. Management emphasized that advanced packaging is intended primarily to ensure and enable wafer demand rather than serve as a standalone profit pool. TSMC is therefore developing multiple packaging architectures, including panel-level packaging, etc, while remaining open to customers using alternative packaging providers, as either approach ultimately supports demand for TSMC wafers.
- **Advanced-packaging execution remains a key differentiator:** TSMC noted that its current advanced-packaging capability supports approximately 5.5x reticle size, with a roadmap to reach approximately 14x reticle sizes within the next several years. While competing roadmaps may also appear attractive, management believes execution and the ability to deliver in high-volume manufacturing will ultimately determine competitive outcomes. TSMC does not believe competitors can use advanced-packaging wins to meaningfully displace its leading-edge wafer business, as foundry and packaging remain fundamentally different businesses.
- **A unified TSMC culture supports overseas manufacturing execution:** TSMC's objective is not to create separate Taiwan and US cultures, but to establish a consistent TSMC culture centered on integrity, commitment, innovation, and customer trust (ICIC). Locally hired Arizona employees spend six to eighteen months training and working in Taiwan, while experienced Taiwan-based employees are assigned to Arizona to work alongside local teams. Over time, the number of Taiwan assignees should decline as locally trained employees assume greater responsibility. TSMC is applying this approach across its overseas manufacturing sites.
- **Management remains comfortable with its 56%-and-above long-term gross margin target:** TSMC reiterated its 6 major drivers of profitability: new-technology ramp, pricing, cost, product mix, utilization, and FX. The N2 ramp is expected to dilute 2026 GM by approximately 2-3ppts, but its profitability should improve over time toward the corporate average; management also noted that N2's structural profitability is better than N3 at a comparable stage. Overseas expansion is expected to dilute GM by 2-3ppts during the early portion of the current five-year period, then widening to 3-ppts later as more overseas fabs begin production. Each new fab will initially create dilution before improving along its learning curve. Taking these puts and takes together, management remains very comfortable that disciplined capacity planning, healthy utilization, cost reduction, productivity gains, and pricing that reflects TSMC's value will enable the company to deliver for shareholders and achieve its long-term gross margin target 'of 56% and higher' with focus on the 'higher' side.

Key Takeaways from Amkor Technology presentation

Presenters: Megan Faust, EVP, CFO & Treasurer, Doug Scott, CVP, Advanced & Mainstream Business Units

- **Arizona capacity expansion:** Amkor discussed its phase 2 of Arizona's Advanced packaging and test campus. Management noted strong demand for advanced packaging, and mentioned that customers are looking for long-term arrangements to secure their supply. Arizona's phase 2 is expected to add 60,000 sq m cleanroom space, in addition to 33,000 sq m added in phase 1. Per initial assessment, revenue from phase 2 will scale around similar levels to phase 1 - \$1 bn revenue with 30% gross margin. Amkor also noted proximity to TSMC's facility in Arizona provides value in terms of quick feedback loop and a shorter cycle time.
- **Strategic areas of growth:** Amkor is focused on its high density fan out and 2.5D products to address the advanced packaging bottleneck. Amkor is also focused on co-packaged optics, and is adding manufacturing capacity to support higher demand levels in both these areas.
- **Communications:** Amkor noted near-term headwinds in the segment driven by (1) memory related demand destruction; (2) capacity shift from Korea to Vietnam and (3) different build cycle from key customers. Amkor expects iOS demand to be relatively better than Android.
- **Compute:** Amkor noted increasing complexity provides more opportunity to the company in packaging. Amkor expects compute revenue to continue to grow at 30% YoY in 2027.
- **Rebalancing factory network:** Amkor discussed its strategic decision to centralize the SIP business, and moving capacity from Korea to Vietnam. Amkor also noted expanding capacity in Portugal, Taiwan, and other areas outside the U.S. to provide geographical flexibility.
- **Long-term financial model:** Amkor discussed its \$11 bn 2030 revenue target, driven by (1) growth in AI and compute business and (2) ramping Arizona phase 1 capacity. Amkor noted phase 2 Arizona capacity will begin producing in 2030 and could present modest upside to the 2030 revenue target. Amkor expects gross margins from Arizona facility to be accretive exiting 2029, into 2030. Amkor expects buybacks to offset dilution from stock compensation, and long-term leverage levels of 1.5X.
- **CapEx:** Amkor noted 2028 CapEx levels to be elevated driven by tool spend for phase 1 and beginning of construction for phase 2. Amkor expects 20%-25% of \$12 bn investment in Arizona in 2028 and 2029, and 10%-15% in 2030 and 2031.

Key Takeaways from NXP presentation

Presenters: Rafael Sotomayor, President & CEO, and Michael Lucarelli, SVP, Investor Relations

- **Demand trends across end markets continue to strengthen, including in automotive:** Industrial and IoT remain the company's best-performing markets year-to-date and have been key growth drivers. In automotive, NXP noted that ~50% of its automotive business consisted of its key growth drivers in 2Q, including SDVs, radar, and connectivity, which are growing at ~20% and where it expects its leadership to drive additional share gains. In China, NXP is focused on system-level solutions to drive leadership, although it does see competition at the component level.
- **Datacenter has become a key growth driver for NXP:** Management discussed NXP's datacenter exposure, noting that increasing networking speeds and rack complexity is intensifying infrastructure requirements, which it sees as a tailwind. NXP struck a bullish tone on the outlook for the segment (over 100% growth expected in 2026), and noted that it is investing in its product portfolio to accelerate growth in this end market.
- **Strategy and pivot to physical AI:** The company is refocusing its portfolio toward physical AI and intelligent systems at the edge, and divesting from areas that are not aligned with that goal. NXP sees itself as a physical AI company with foundations in automotive, and emphasized its focus on keeping pace with innovation in robotics where it sees several opportunities across battery management, reflex/reaction, and edge compute.
- **Broad inventory re-stocking has not materialized, although visibility has improved:** NXP sees positive demand signals as evidenced by its growing backlog and expanding lead times, which is improving its revenue visibility. While its customer conversations have been constructive, management noted that OEM inventories remain lean, with little evidence thus far of broad-based inventory restocking.
- **Price increases expected to have a cumulative effect on revenue over time:** NXP has selectively increased prices in a staggered and gradual manner, helping to offset higher input costs that cannot be mitigated with operational measures. NXP expects the effect on its topline to accumulate over the next few quarters.
- **Hybrid manufacturing strategy provides capacity and cost upside:** Management discussed NXP's JVs (VSMC and ESMC), its shift to 300 mm from 200 mm wafers, and the importance of its hybrid manufacturing model in maintaining supply-side flexibility while remaining cost competitive ahead of a potential broad cyclical upswing in analog.

Key Takeaways from Cadence presentation

Presenters: Anirudh Devgan, Ph.D., President & CEO

- **Chip design activity remains robust:** Cadence continues to see increased levels of chip design activity from traditional semi companies as well non-traditional customers (i.e. hyperscalers and system companies).

- **Customer adoption of AI offerings continues to grow:** Cadence's AI portfolio continues to broaden in scope and customers, and exposes the company to a new TAM of AI agents. Cadence noted better interoperability between its own agents in the top layer and tools in the middle layer as a key source of differentiation, vs. customers doing their own top layer (agents) to work with Cadence's tools (middle layer).
- **Agentic AI solutions to be a key driver of growth:** Cadence discussed a new business model of consumption + subscription for the top layer (agents), while continuing with subscription revenue for the middle layer (tools). Cadence noted chip complexity is expected to grow by 48X over the next five years, which is a key driver for its agentic AI solutions – enabling accelerated recurring revenue growth.
- **Hardware business:** Cadence highlighted strong momentum in its Hardware business, driven by growing adoption of hardware-accelerated design verification as customers seek to shorten development cycles amid increasing chip design complexity and rising compute requirements.
- **System design & analysis:** Cadence noted SDA algorithms are similar to EDA algorithms, which makes it a natural extension of Cadence's core strength of computational software.
- **Physical AI:** Cadence shared optimism around physical AI, and expects it to be a large market opportunity in the medium-term. Cadence noted its Hexagon acquisition to pursue this physical AI opportunity. Cadence also expects more silicon content to be driven by physical AI, which should also benefit its traditional EDA business.

Key Takeaways from Sandisk presentation

Presenters: David Goeckeler, CEO, and Luis Visoso, CFO

- **Business model shift to NBMs:** SanDisk continues to see a shift in the NAND market away from a short-term spot pricing model to one dominated by longer-term agreements, enabled in part by strong and growing datacenter demand. Management also discussed the key financial elements underpinning its NBMs, including the floor pricing component which enables a gross margin of ~80% for the majority of the business in a downside scenario.
- **Confidence in long-term financial targets:** Management sees increased revenue visibility into FY27/FY28, underpinned by 50%/67% of its planned volume covered by NBMs. SanDisk sees NAND supply growth as likely to remain muted for the foreseeable future, against a backdrop of continued demand increases driven by the proliferation of AI inference. Management also sees supply additions from competitors in China as mainly being absorbed by the local market.
- **Long-term optionality in HBF and KV Cache:** Management is optimistic about the long term outlook for both HBF – which it sees as a potential solution to the “memory wall” in AI through the higher density it provides. Management previously described KV Cache as critical for AI inference, representing ~35% of the 1.2 ZB AI

datacenter memory TAM by 2032.

- **Low capital intensity remains a key differentiator:** SanDisk recently extended its JV with Kioxia through 2034, and highlighted that the arrangement, coupled with its IP ownership and R&D investments, has enabled the JV's efficient manufacturing capabilities and disproportionate share of bit growth relative to CapEx. Management added that it has line of sight through its BiCS platform for several years which can support further bit growth at the same low capital intensity (~5%).
- **Capital returns centered on share repurchases for now:** SanDisk reiterated that it views share repurchases as the primary vehicle for returning excess capital to shareholders, adding that it has already executed ~\$4.5 bn in buybacks. Management also added that it remained open-minded to dividends in the future.

Key Takeaways from IBM presentation

Presenters: Jim Kavanaugh, CFO

- **Software.** Following its miss in 2Q, IBM noted that a re-prioritization of CapEx spending led to a shortfall in Software – although this only impacted ~15% of the portfolio. The company expects to grow its Software segment 6% – 8% in 2026, driven by Red Hat momentum and double-digit growth in its high value recurring revenue in 2H26. In the long run, IBM expects growth approaching low-double digits (~10%) in Software, primarily led by strength in Red Hat, GenAI demand, and continuous M&A synergies.
- **Consulting.** IBM says it sees healthy signings in its Consulting business which it believes differentiates it from its competitors in the space, where it sees traditional consulting as being completely disrupted by agentic AI. Management highlighted the synergistic value of its status as a platform company and the resulting revenue multiplying effect, and noted that 20% of its new Software revenue was driven by its Consulting business.
- **Generative AI.** IBM believes enterprise AI adoption remains in the early stages, and sees itself as uniquely well positioned to address clients' adoption pain points around AI deployment complexity. IBM highlighted the benefits of its cost-effective, open-source AI tools such as its smaller LLMs, RHEL AI, WatsonX code generation tools, and WatsonX Orchestrate for scaling agentic AI. Management also highlighted the growing importance of customer's data in the context of AI, and sees GenAI as a key new source of value in its Consulting business.
- **Infrastructure.** Following its mainframe revenue miss in 2Q, IBM noted that it has closed ~40% of its previously deferred deals. IBM's mainframe offering remains strong, led by performance from the z17 mainframe which it described as the most successful program in IBM history with 130% of revenue expansion vs. the prior program.
- **Quantum:** IBM noted continued progress on its quantum program and highlighted its substantial scale relative to the rest of the industry. The company is building the first quantum wafer foundry (Anderon) with support from the US Department of

Commerce, and is on track to demonstrate the first error-corrected quantum computer by 2028. Management sees a multi-billion dollar commercial opportunity before the end of the decade, and a total value-creation TAM of \$850 bn over time, adding that IT providers can capture 20%-40% of that opportunity.

- **Financial/M&A.** Following its recent guide down, IBM remains confident in its ability to deliver 4% - 5% revenue growth in CC for 2026, while increasing free cash flow by ~\$1bn YoY and delivering 100bps of pre-tax margin expansion. On the M&A side, the company noted it remains focused on identifying assets that align with its areas of strength across hybrid cloud, GenAI, and Quantum, while presenting attractive return profiles.

Key Takeaways from Teradyne presentation

Presenters: Greg Smith, President & CEO

- **Key growth drivers:** Teradyne highlighted its exposure across the AI infrastructure stack, with semiconductor test (~80% of revenue), robotics (~10%), and product test (~10%) serving complementary needs from wafer and package testing through system-level validation. The company sees increasing opportunities with hyperscale customers, whose diverse requirements across chip, system, and datacenter testing create reinforcing demand across all three businesses.
- **ATE growth:** Management expects test equipment market to at least grow in line with overall semiconductor CapEx over the next 3-5 years, representing ~8% of overall semiconductor CapEx (vs. ~4% historically) as test intensity continues to move up.
- **Compute:** Teradyne expects AI-driven growth to accelerate through 2027, supported by rising share gains at a major merchant compute customer, expanding engagements with hyperscale custom silicon customers, and strong momentum in ASIC testing. Teradyne expects meaningful production share gains at a second hyperscale cloud compute customer in 2027, complementing its path to ~30% share at a major merchant compute customer over the next 3-5 years. Management sees custom AI silicon as a larger growth opportunity than networking, with multiple customer programs representing hundreds of millions to over \$1 billion of annual TAM.
- **Co-packaged optics:** Teradyne views CPO testing as a significant long-term opportunity, with its test equipment TAM expected to grow from roughly \$100mn today to \$300mn-\$700mn by 2028 and potentially ~\$1bn by 2030. While the market remains nascent and highly concentrated, management expects the competitive landscape to evolve as increasing lane counts, data rates, and device-handling complexity drive a more mature test market where Teradyne and Advantest compete for share.
- **Memory:** Mr. Smith expects the ongoing momentum in the memory testing market to continue, with modest growth in HBM, strong growth in DRAM and NAND in 2027.
- **Robotics:** Teradyne highlighted strong momentum in Robotics, with outsized

revenue growth expected in 2026 and 2027. The company expects robotics-assisted manufacturing in datacenters to be the strongest growing area in robotics. Management also noted growing engagement with a marquee customer as it explores additional workflows.

- **Gross margins and OpEx:** Teradyne expects gross margins to remain in the current 59%-60% range. Teradyne expects significant operating leverage as it grows in size in the future, but is leaning into OpEx currently to pursue additional growth opportunities. Teradyne still expects revenue to grow faster than OpEx.

Key Takeaways from Applied Materials presentation

Presenters: Gary Dickerson, President & CEO

- **Strong customer order visibility and growing demand:** Applied Materials highlighted AI and datacenter as the primary drivers of semiconductor spending, with leading-edge logic, advanced packaging, and memory accounting for ~80% of WFE in 2026 and similar mix in 2027. Applied Materials reiterated its outlook for ~40% revenue growth in 2026, and expects strong growth to continue into 2027 and extend into 2028. Management noted that agentic AI is driving stronger investment across CPUs and DRAM, while customer demand visibility remains robust with rolling 8-quarter forecasts, long-term commitments extending to 2030, and continued upward revisions to capacity plans.
- **AI and product innovation:** Mr. Dickerson noted shorter time to market, and yield and output innovation driven by AI as key ROI provider in the industry to Applied and to its customers. Applied also cited improved value proposition in services using AI as an additional driver of ROI in the industry.
- **Industry positioning across key end-markets:** Applied Materials highlighted its leadership across leading-edge logic, memory, and advanced packaging, with advanced packaging revenue expected to grow ~70% YoY in 2026 to a \$2.5-\$3.0bn revenue run-rate. Management also noted new fab locations as a tailwind to services as customers expand their geographical footprints.
- **ICAPS:** Applied Materials expects China revenue to grow YoY in 2026 with modest share gains, supported by strength in the ICAPS market. Longer term, management sees ICAPS sustaining a mid-to-high single digit CAGR, with acceleration in 2027 versus 2026.
- **Process control:** Applied Materials expressed strong confidence in its Process Control business, which is expected to grow more than 50% in 2026, driven by increasing demand for yield improvement and materials engineering at advanced nodes. Management highlighted leadership in e-beam technology, imaging, and materials analysis, positioning Process Control as a durable high-growth business for many years.
- **EPIC:** Mr. Dickerson is pursuing high-velocity co-innovation with customers through EPIC. This facility will help Applied progress on technologies up to four nodes in advance. For customers, it helps them to bring new products faster to the market.

Key Takeaways from SiTime presentation

Presenters: Rajesh Vashist, Chairman & CEO, and Beth Howe, EVP, Finance & CFO

Bottom Line: We have three key takeaways: (1) Datacenter remains the key growth driver for the company, driven by AI server units, increasing market share, and higher content; (2) Renesas IC clock revenue growth is expected to inflect in 2027; (3) revenue growth is likely to exceed the company's long term targets over the next few quarters.

- **Datacenter outlook:** SiTime continues to see strong momentum in its datacenter business, driven by higher content requirements for AI inference, increasing market share, and the ongoing ramp of higher-speed optical networking products at 1.6 Tbps (where it expects its market share to double to 80%). Management sees content per rack as continuing to increase, to \$400-\$600, and potentially up to \$1,000 in certain configurations. It also highlighted design wins across GPUs, CPUs, ASICs, and switches, where it estimates timing content of between \$20-\$25 per switch.
- **Renasas IC clock portfolio integration:** The recently acquired portfolio of IC clocks from Renesas, which management sees as complimentary to its oscillator business, remains supply-constrained. However, a revenue inflection could occur in mid-2027 as the company manages supply and begins to benefit from cross-selling synergies with its existing products.
- **Consumer and IoT:** Although typical seasonality patterns have been somewhat disrupted recently, SiTime expects a seasonal slowdown in the mobile/consumer business in 1Q27. Management also highlighted design wins in other areas, and struck a bullish tone overall.
- **Automobile and Industrial:** Management highlighted several opportunities across different end markets, including in military/aerospace, which it says could reach \$100mn in annual revenue in the near term and could double for the next two years. SiTime also highlighted smart grids, where it sees an incremental SAM of ~\$200 mn, and physical AI, where the company expects a gradual buildup in revenue before an inflection, which it added could occur in 2028.
- **OpEx increases likely to be outpaced by revenue growth:** Following the integration of the ~160 Renesas engineers from its recent acquisition and plans to grow the sales force, SiTime expects its operating expenses to continue to grow in the medium term, albeit at a lower pace than revenue. Management noted that its revenue growth may exceed its long-term targets over the next several quarters and reiterated its prior guidance of ~68% gross margin in the current quarter with an operating margin of 34% or higher.
- **Capital structure and allocation:** Management discussed its decision to pursue the convertible bond market to help finance its recent acquisition, but noted that it expects its cash flow generation to help lower net debt under 2x within 24 months or sooner. On capital allocation, the company said it remains focused on prioritizing reinvestment in the business and growth over share repurchases.

Valuation/Risks

Amkor (Neutral): Our 12-month price target is \$65, which is based on a 25X P/E multiple applied to our normalized EPS of \$2.60. Key upside/downside risks: (1) better-than-expected volumes in the 2.5D packaging business; (2) better gross margin leverage from utilization of new production facilities; (3) failure to execute on the 2.5D packaging roadmap; (4) failure to execute on new smartphone design wins.

NXP (Buy): Our 12-month price target is \$325, which is based on an 18X P/E multiple applied to our normalized EPS forecast of \$18.00. Key downside risks: (1) renewed end-demand weakness in Automotive; (2) timing and the shape of the recovery remain uncertain; (3) unfavorable pricing.

Cadence (Buy): Our 12-month target price is \$470, which is based on a 45.5X multiple applied to our normalized EPS estimate of \$10.50. Downside risks: (1) export restrictions; (2) market share losses; (3) fewer custom chip designs.

Sandisk (Buy): Our 12-month target price is \$2,200, which is based on a 20X P/E multiple applied to our normalized EPS estimate of \$110. Key downside risks include: (1) Long-term structural change in NAND pricing fails to materialize; (2) YMTC continues to iterate on its roadmap; (3) SanDisk fails to gain eSSD traction.

IBM (Buy): Our 12-month price target is \$270, which is based on 20X our one-year forward earnings estimates. Downside risks: 1) increased macro headwinds, 2) slowdown in consulting, 3) AI bookings deceleration, 4) dilutive M&A.

Teradyne (Buy): Our 12-month price target is \$465, which is based on a 37X P/E multiple applied to our normalized EPS forecast of \$12.50. Key downside risks: (1) delays in the revenue ramp with a merchant GPU customer, (2) significant quarterly volatility due to the lumpy nature of large AI-related infrastructure projects, (3) slower-than-expected recovery in Mobile.

Applied Materials (Buy; on CL): Our 12-month price target is \$670, which is based on a 32X P/E multiple applied to our normalized EPS estimate of \$21.00. Key downside risks include: (1) incremental export restrictions; (2) local China-based supplier share gains.

SiTime (Buy): Our 12-month price target is \$900, which is based on 60X applied to our normalized EPS forecast of \$15.00. Key downside risks include: (1) any incremental slowing the pace of AI datacenter spending; (2) design win momentum at key datacenter or smartphone customers.

TSMC (Buy): Our 12-month price target is NT\$3,100, which is derived by applying a target P/E multiple of 22x to our 2027E EPS. Our 22x target P/E is benchmarked against 1.0stdv above its 5-year trading average. For the ADR (TSM), our 12-month price target is US\$620, which is based on a USD/TWD rate of 30.0 and an ADR premium of 20%. Key downside risks include: (1) further deterioration in end-demand recovery impacting capacity utilization; (2) slower customer node migrations; (3) slowdown in AI investment resulting in lower long-term semiconductor content growth; (4) poor yields/execution resulting in worse-than-expected profitability; (5) stronger competition resulting in ASP/profitability erosion; and (6) unfavorable FX trend or higher-than-expected cost increase weighing on the margin outlook.

Disclosure Appendix

Reg AC

We, James Schneider, Ph.D., Evelyn Yu, Anmol Makkar, Luya You, Khalil Fenina and Marshall Wong, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Disclosures

Rating and pricing information

Amkor Technology Inc. (Neutral, \$51.35), Applied Materials Inc. (Buy, \$468.85), International Business Machines Corp. (Buy, \$239.94), NXP Semiconductors NV (Buy, \$223.32), SanDisk Corp. (Buy, \$1,764.17), SiTime Corp. (Buy, \$589.87), TSMC (Buy, NT\$2,465.00), TSMC (ADR) (Buy, \$435.36) and Teradyne Inc. (Buy, \$383.69)

The rating(s) for Amkor Technology Inc., Applied Materials Inc., Cadence Design Systems Inc., International Business Machines Corp., NXP Semiconductors NV, SanDisk Corp., SiTime Corp. and Teradyne Inc. is/are relative to the other companies in its/their coverage universe: ARM Holdings, Accenture Plc, Advanced Micro Devices Inc., Amkor Technology Inc., Analog Devices Inc., Applied Materials Inc., Broadcom Inc., Cadence Design Systems Inc., Camtek, Cognizant Technology Solutions, Credo Technology Group, EPAM Systems Inc., Entegris Inc., GlobalFoundries Inc., Globant SA, Intel Corp., International Business Machines Corp., KLA Corp., Lam Research Corp., MKS Instruments Inc., Marvell Technology Inc., Microchip Technology Inc., Micron Technology Inc., NXP Semiconductors NV, Nvidia Corp., ON Semiconductor Corp., Onto Innovation, Qnity, Qualcomm Inc., SanDisk Corp., Seagate Technology, SiTime Corp., Synopsys Inc., TaskUs Inc., Teradyne Inc., Texas Instruments Inc., Western Digital Corp.

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Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
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Price target and rating history chart(s)

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Target price history table(s)

SiTime Corp. (SITM)

Date of report	Target price (\$)	Closing price (\$)
06-Aug-26	900.00	687.50
06-May-26	800.00	623.33
05-Feb-26	435.00	410.29
15-Dec-25	420.00	363.25

TSMC (2330.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
16-Jul-26	3,100.00	2,470.00
02-Jul-26	3,000.00	2,465.00
07-Apr-26	2,750.00	1,860.00
15-Jan-26	2,600.00	1,690.00
04-Jan-26	2,330.00	1,585.00
16-Oct-25	1,720.00	1,485.00
30-Sep-25	1,600.00	1,305.00
17-Jul-25	1,370.00	1,130.00
25-Jun-25	1,210.00	1,070.00
22-May-25	1,145.00	982.00
17-Apr-25	1,190.00	847.00
06-Apr-25	1,200.00	942.00
16-Jan-25	1,400.00	1,105.00
03-Jan-25	1,355.00	1,075.00
17-Oct-24	1,320.00	1,035.00
30-Sep-24	1,270.00	957.00
18-Jul-24	1,230.00	1,005.00
18-Jun-24	1,160.00	943.00
21-Mar-24	975.00	784.00
18-Jan-24	760.00	588.00
02-Jan-24	758.00	593.00
19-Oct-23	725.00	546.00
28-Sep-23	685.00	523.00
14-Sep-23	668.00	550.00

Cadence Design Systems Inc. (CDNS)

Date of report	Target price (\$)	Closing price (\$)
13-Jul-26	470.00	377.92
27-Oct-25	410.00	351.40
29-Jul-25	400.00	366.26
10-Jul-25	380.00	322.66

Teradyne Inc. (TER)

Date of report	Target price (\$)	Closing price (\$)
05-Jul-26	465.00	369.09
30-Apr-26	350.00	343.47
03-Feb-26	300.00	282.98
15-Dec-25	230.00	194.70
29-Oct-25	148.00	173.94
08-Oct-25	130.00	144.60
30-Jul-25	85.00	107.65
10-Jul-25	80.00	98.62
30-Jan-25	140.00	115.08
26-Jul-24	151.00	126.46
20-May-24	161.00	139.51
26-Apr-24	118.00	114.13
01-Feb-24	107.00	95.14
26-Oct-23	103.00	85.12

TSMC (ADR) (TSM)

Date of report	Target price (\$)	Closing price (\$)
16-Jul-26	620.00	409.74
02-Jul-26	600.00	434.16
07-Apr-26	550.00	345.32
15-Jan-26	520.00	341.64
04-Jan-26	466.00	319.61
16-Oct-25	344.00	299.84

Applied Materials Inc. (AMAT)

Date of report	Target price (\$)	Closing price (\$)
13-Aug-26	670.00	534.54
05-Jul-26	645.00	603.04
15-May-26	520.00	436.62
13-Feb-26	390.00	354.91
09-Jan-26	310.00	301.18
08-Oct-25	250.00	217.51

TSMC (ADR) (TSM)

Date of report	Target price (\$)	Closing price (\$)
30-Sep-25	320.00	279.29
17-Jul-25	274.00	245.60
25-Jun-25	242.00	222.74
22-May-25	229.00	196.19
17-Apr-25	219.00	151.74
06-Apr-25	222.00	146.80
16-Jan-25	259.00	214.79
03-Jan-25	254.00	208.61
17-Oct-24	248.00	205.84
30-Sep-24	234.00	173.67
18-Jul-24	227.00	171.87
18-Jun-24	218.00	179.69
18-Apr-24	183.00	132.27
21-Mar-24	188.00	139.45
18-Jan-24	123.00	113.03
02-Jan-24	122.00	101.53
19-Oct-23	115.00	92.91
28-Sep-23	110.00	86.41
14-Sep-23	109.00	91.47

Applied Materials Inc. (AMAT)

Date of report	Target price (\$)	Closing price (\$)
15-Aug-25	215.00	161.76
10-Jul-25	225.00	198.03
10-Jan-25	202.00	171.65
15-Nov-24	230.00	168.88
02-Oct-24	228.00	201.15
17-May-24	240.00	212.08
16-Feb-24	220.00	199.57
17-Nov-23	156.00	148.59
19-Sep-23	152.00	137.71

International Business Machines Corp. (IBM)

Date of report	Target price (\$)	Closing price (\$)
23-Jul-26	270.00	206.65
23-Apr-26	335.00	231.08
29-Jan-26	365.00	309.24
08-Oct-25	350.00	289.46
24-Jul-25	320.00	260.51
01-Jul-25	310.00	291.20
06-Apr-25	270.00	227.48
30-Jan-25	275.00	258.27
30-Sep-24	250.00	221.08
07-Aug-24	220.00	186.80
25-Jul-24	205.00	191.98
24-Jun-24	200.00	175.01

Amkor Technology Inc. (AMKR)

Date of report	Target price (\$)	Closing price (\$)
28-Apr-26	65.00	71.36
10-Feb-26	43.00	53.46
09-Jan-26	38.00	52.29
28-Oct-25	29.00	32.16
08-Oct-25	26.00	30.61
29-Jul-25	22.00	25.08
10-Jul-25	20.00	21.89
11-Feb-25	30.00	21.60
29-Oct-24	32.00	27.98
30-Apr-24	36.00	32.35
29-Feb-24	32.00	31.02

SanDisk Corp. (SNDK)

Date of report	Target price (\$)	Closing price (\$)
05-Jul-26	2,200.00	1,745.00
01-May-26	1,200.00	1,187.00
30-Jan-26	700.00	576.25
09-Jan-26	320.00	377.41
07-Nov-25	280.00	239.48
08-Oct-25	140.00	131.88
10-Jul-25	55.00	46.95

NXP Semiconductors NV (NXPI)

Date of report	Target price (\$)	Closing price (\$)
29-Apr-26	325.00	289.25
10-Jul-25	276.00	233.19
04-Feb-25	260.00	202.40
10-Jan-25	257.00	206.75
05-Nov-24	260.00	224.65
07-Oct-24	277.00	234.02
23-Jul-24	285.00	262.30
30-Apr-24	275.00	256.19
06-Feb-24	240.00	223.00
07-Nov-23	226.00	185.80

Price targets shown in table(s) are unadjusted for corporate actions.

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